

KNOXVILLE TYSON, TN, USA

WMO: 723260

Lat: 35.818N

Lon: 83.986W

Elev: 962

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.4	21.0	3.6	6.9	20.7	7.9	8.6	24.9	25.4	55.3	22.9	52.5	5.7	40	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	92.7	73.6	90.4	73.3	88.4	72.7	76.9	87.2	75.9	85.5	75.0	84.2	7.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	131.2	81.2	73.0	127.0	80.2	72.1	123.2	79.4	40.8	87.3	39.9	86.0	39.0	84.4	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	18.2	15.8	DB	9.9	95.2	6.2	3.4	5.4	97.6	1.8	99.6	-1.6	101.5	-6.1	104.0
			WB	8.6	78.5	5.8	1.3	4.4	79.5	1.0	80.3	-2.2	81.0	-6.4	82.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	38.8	43.1	50.8	60.0	68.2	75.4	78.2	77.8	72.0	60.6	49.3	42.2
	DBStd	15.98	10.78	9.96	9.53	8.17	6.79	4.48	3.64	3.72	5.77	7.92	8.47	9.24
	HDD50	1134	375	234	110	15	0	0	0	0	0	11	113	276
	HDD65	3486	812	614	447	189	49	2	0	0	13	180	474	706
	CDD50	4709	28	41	134	315	564	763	875	862	659	340	92	36
	CDD65	1585	0	1	6	39	148	315	410	397	223	43	3	0
	CDH74	13424	0	6	47	333	1173	2651	3608	3506	1757	326	16	1
	CDH80	4883	0	0	3	46	311	976	1439	1422	622	63	1	0
(14) Wind	WSAvg	5.9	6.7	7.2	7.3	7.3	6.3	5.6	5.4	4.6	4.6	4.9	5.3	6.0
(15) Precipitation	PrecAvg	48.8	4.4	4.1	5.0	4.1	4.1	4.1	5.0	3.3	3.3	2.8	3.9	4.8
	PrecMax	69.3	12.7	8.8	11.8	11.1	11.0	9.4	12.7	6.7	9.2	6.0	7.6	11.6
	PrecMin	32.5	0.9	0.7	1.7	0.4	0.7	0.5	0.3	0.9	0.2	0.0	1.1	0.4
	PrecStd	8.2	2.0	1.9	2.4	2.1	2.1	2.0	2.4	1.5	2.0	1.5	1.5	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.1	73.2	78.9	84.2	89.2	93.7	95.2	95.7	93.0	85.4	76.2	70.8
		MCWB	59.7	61.6	62.7	65.9	70.6	73.6	74.7	74.2	70.3	69.4	64.1	62.1
	2%	DB	63.8	67.9	74.5	81.1	86.2	90.7	92.2	92.4	89.7	81.2	71.3	65.0
		MCWB	57.3	58.6	60.1	64.3	69.3	73.3	74.7	73.8	70.5	66.9	60.8	58.2
	5%	DB	59.6	63.4	70.5	78.2	83.6	88.2	89.7	89.9	86.8	77.5	67.4	60.9
		MCWB	53.9	54.3	58.0	62.8	68.5	72.3	74.3	73.6	70.4	65.1	58.8	55.9
	10%	DB	55.1	58.4	66.6	74.6	80.9	85.8	87.5	87.5	83.6	74.1	63.8	56.9
		MCWB	50.1	50.9	55.7	61.3	67.7	71.3	73.6	73.0	69.4	63.5	56.4	52.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	63.8	64.8	69.4	73.9	77.4	78.4	78.0	75.2	73.2	67.1	63.6
		MCDB	66.9	70.4	74.9	81.0	83.2	88.3	88.8	88.8	85.0	80.6	71.9	68.5
	2%	WB	58.3	60.4	62.4	67.0	72.0	75.7	77.1	76.6	73.5	70.0	63.7	59.8
		MCDB	63.1	66.6	71.2	77.0	82.0	85.9	87.2	87.1	83.0	76.8	68.7	63.5
	5%	WB	55.1	56.5	60.1	64.9	70.6	74.6	76.1	75.5	72.3	67.7	60.6	56.8
		MCDB	58.9	61.4	67.7	74.1	80.2	84.1	85.6	85.2	81.0	74.4	65.7	60.2
	10%	WB	51.1	52.6	57.6	63.1	69.1	73.3	75.2	74.6	71.2	65.4	57.3	52.9
		MCDB	54.0	57.1	64.5	71.3	77.6	82.2	84.0	83.7	79.2	71.6	62.4	56.3
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.8	19.8	21.1	20.0	18.8	17.7	18.6	19.6	20.7	19.9	16.7
		MCDBR	20.3	23.6	24.9	24.8	22.4	21.7	20.5	21.5	23.3	23.6	24.3	21.6
	5% WB	MCWBR	14.6	15.4	13.8	11.8	8.9	7.5	6.9	6.9	7.7	10.7	14.5	15.2
		MCWBR	17.5	20.3	20.5	20.2	19.4	18.6	17.8	18.4	18.7	18.5	19.4	18.6
(40) Clear-Sky Solar Irradiance	taub		0.311	0.321	0.352	0.403	0.455	0.490	0.523	0.530	0.423	0.368	0.331	0.311
		taud	2.507	2.475	2.418	2.273	2.154	2.092	2.026	1.984	2.270	2.420	2.481	2.531
	Ebn at Noon		283	292	291	279	265	255	246	240	264	271	272	276
		Edh at Noon	26	30	35	42	48	51	54	56	40	31	26	24
(44) All-Sky Solar Radiation	RadAvg		731	910	1279	1637	1813	1930	1826	1713	1470	1178	857	634
	RadStd		59	112	113	113	130	127	118	87	132	141	92	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	+241	+164
(47) Regional (26 neighbors)		+0.97	N/A	N/A	N/A	N/A	N/A	N/A	+265	+166

Nomenclature: See separate page